Engineering Fracture Mechanics 234 (2020) 107109

Contents lists available at ScienceDirect

Engineering Fracture Mechanics

journal homepage: www.elsevier.com/locate/engfracmech

Fracture behavior and damage mechanisms of sandstone subjected to wetting-drying cycles

Xin Cai[a], Zilong Zhou[a,*], Lihai Tan[a], Haizhi Zang[a,b], Zhengyang Song[c,*]

[a] School of Resources and Safety Engineering, Central South University, Changsha, Hunan 410010, PR China
[b] Faculty of Engineering and Built Environment, ARC Centre of Excellence for Geotechnical Science and Engineering, The University of Newcastle, Callaghan 2308, Australia
[c] Geotechnical Institute, TU Bergakademie Freiberg, Gustav-Zeuner-Straße 1, Freiberg 09599, Germany

ARTICLE INFO

Keywords:
Sandstone
Cyclic wetting-drying
SHPB
NSCB
Fracture toughness

ABSTRACT

Rocks are usually subjected to cyclic wetting-drying due to the periodical changes in moist condition. To investigate the effect of wetting-drying cycles on the fracture behavior of sandstone, notched semi-circular bending tests were performed on sandstone samples after every ten cycles (a total of 50 cycles) under both quasi-static and dynamic loading conditions. Tests results indicate that both of fracture toughness and energy dissipation of sandstone significantly decrease with the increase of cycle number. Moreover, the microstructural characteristics reveal that after cyclic wetting-drying treatments, the calcite cementation between grains gradually dissolves and the density and width of inter-granular fracture increase, which could be responsible for the deterioration of rock mechanical properties.

1. Introduction

For many projects related to rock engineering, such as slope, dam, mining, tunneling and highway subgrade, the rock masses commonly suffer cyclic wetting-drying due to the periodical variations of water content in moisture conditions [1–5]. As a crucial part of weathering process, the effect of wetting-drying cycles can aggravate rock weathering, weaken rock properties, which would trigger a large number of engineering geological disasters, such as landslides, gob collapse, debris flow and ground subsidence. Therefore, a comprehensive and in-depth understanding of the effects of wetting-drying cycles on rock properties is significant for the design, construction and maintenance of rock engineering in humid regions.

In recent years, a great effort has been made towards investigating the variation in the physical properties and mechanical behavior of rock subjected to wetting-drying cycles [1–3,6–21]. The majority of researches are typically interested in the uniaxial compressive [2,7–10,14,16,18–21] and tensile strength [3,9,11,12,22]. For instance, Hale and Shakoor [7] conducted uniaxial compression tests on six types of sandstone subjected to 50 wetting-drying cycles. It was documented that the wetting-drying processes within 50 cycles do not show any obvious significances with respect to the UCS of sandstones. Özbek [18] investigated the variations of UCS for Kayseri ignimbrites after the treatments of cyclic wetting-drying. It was found that the UCS of rock sample undergoing 50 cycles decreases by 11.40–16.08%. Khanlari and Abdilor [21] stated that observable macrocracks appear in rock surface after 40 wetting-drying cycles, and the UCS is reduced by 11–21%. Huang et al. [14] reported that the loss of UCS and elastic modulus shows a logarithmic manner with the number of wetting-drying cycles. They also found that compared with the mechanical properties of sandstone, the mudstone are more sensitive in terms of properties deterioration when exposed to cyclic wetting and

* Corresponding authors.
E-mail addresses: xincai@csu.edu.cn (X. Cai), zlzhou@csu.edu.cn (Z. Zhou), villasoon@126.com (Z. Song).

https://doi.org/10.1016/j.engfracmech.2020.107109
Received 30 April 2020; Received in revised form 15 May 2020; Accepted 16 May 2020
Available online 24 May 2020
0013-7944/ © 2020 Elsevier Ltd. All rights reserved.

Nomenclature

UCS	Uniaxial Compressive Strength
CCBD	Central Cracked Brazilian Disc
SEM	Scanning Electron Microscopy
SHPB	Split Hopkinson Pressure Bar
ISRM	International Society for Rock Mechanics and Rock Engineering
SR	Short Rod
CCNBD	Cracked Chevron Brazilian Disc
CB	Chevron Bending
NSCB	Notched Semi-Circular Bending
SIF	Stress Intensity Factor
K_{IC}^{st}	quasi-static fracture toughness
K_{IC}^{dy}	dynamic fracture toughness
K_I^{dy}	dynamic SIF
$\dot{K}_I$	loading rate
S	distance between the two supporting pins
l_n	notch length of NSCB sample
R	radius of sample
B	thickness of sample
Y'	non-dimensional SIF
$\bar{F}$	average dynamic force
F_{max}	maximum value of force
A_b	cross-sectional area of bars
E_b	elastic modulus of bars
C_b	P-wave velocity of bars
R^2	correlation coefficient
W_s	energy dissipation
W_I	incident energy
W_R	reflected energy
W_T	transmitted energy
σ_{sp}	swelling pressure
K_{sp}	SIF induced by swelling pressure
r	pore radius
l	length of subcritical crack
K_{th}	SIF threshold for subcritical cracking
E_c	elastic modulus of smectite
ε_c	swelling strain of smectite
ν_c	poisson ratio of smectite

drying. Yao et al. [9] found that the UCS and elastic modulus of sandstone decrease with the increasing number of wetting-drying cycles. They pointed out that the content of calcite decreases after cyclic treatments and suggested that the calcite dissolution is the primary cause of sandstone deterioration. For tensile strength, Zhao et al. [11] performed Brazilian splitting tests on the sandstone containing less than 2% clay minerals. Only slight reduction (about 5%) in the tensile strength was observed after 15 cycles. Nevertheless, Wen et al. [22] reported that the tensile strength of the sandstone with higher clay content decreases up to 43.5% when subjected to 7 wetting-drying cycles.

Besides above-mentioned compressive and tensile properties, the fracture toughness, reflecting the resistance of rock for crack initiation and propagation, is another important parameter for rock fragmentation [23]. For current, quite limited investigation related to the impact of cyclic wetting-drying on fracture toughness of rock materials can be referred to. Wen et al. [22,24,25] studied the fracture behavior of sandstone exposed to wetting-drying cycles by using central cracked Brazilian disc (CCBD) methods. They concluded that both the fracture toughness and crack propagation radius decrease with the increase of cycle number. In addition, the degradation of fracture toughness is always more pronounced than that of tensile strength. However, these studies are limited to quasi-static loading. For in-situ condition, the rock structures are usually failed by dynamic disturbance, such as shock, explosion and seismic events [26–28]. The dynamic behavior of rock materials is very distinct from the quasi-static one [29]. So far, the dynamic fracture behavior of rock after cyclic wetting and drying treatments have been not generally investigated yet.

For our work, the aim is to understand the deterioration of rock properties induced by cyclic wetting and drying. Some physical parameters including density, porosity and P-wave velocity were monitored after every ten wetting-drying cycles (a total of 50 cycles is adopted). The evolution of micro structures in rock sample induced by wetting-drying cycles were observed by using scanning electron microscopy (SEM). Dynamic notched semi-circular bending (NSCB) tests were carried out on sandstone samples by split Hopkinson pressure bar (SHPB) setup to obtain the dynamic fracture toughness and energy dissipation. The quasi-static fracture toughness was also measured for comparsion.

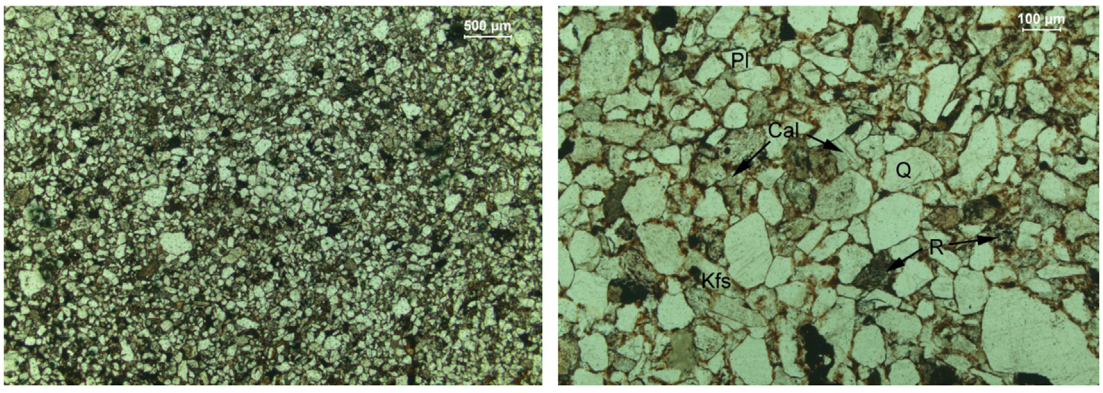


Fig. 1. Optical photomicrographs of the KM sandstone with (a) 20 and (b) 100 magnifications.

2. Experimental methodology

2.1. Material characterization

In this study, a red sandstone collected from an open quarry at Southeast of Kunming (KM) city (Yunnan province, China) was used as the test material due to its homogeneous and isotropic characters. The microstructure of the KM sandstone (Fig. 1) shows that it is fine-grained with an average grain size of 65 μm, and the radius of mineral grain ranges between 5 μm and 240 μm. Grains are primarily cemented by calcite and hematite minerals within pores. The pore radius varies between 5 μm to 100 μm. The mineralogical content of the KM sandstone was quantified by using the Energy Dispersive X-ray spectrometer (EDX MLA250). The EDX analysis results indicate that the main constituent minerals of the KM sandstone contain 42.74% of quartz, 16.85% of omphacite, 13.9% of calcite, 11.19% of hematite, 9.08% of feldspar and 4.98% of clay minerals (dominantly smectite).

2.2. Test method and sample preparation

International Society for Rock Mechanics and Rock Engineering (ISRM) has recommended four standardized methods to accurately measure the mode I fracture toughness of rock materials. These methods can be categorized two groups, i.e., compression-based type including short rod (SR) method [30] and cracked chevron Brazilian disc (CCNBD) method [31], and bending-based type containing chevron bending (CB) method [30] and notched semi-circular bending (NSCB) method [23,32]. These methods have been extended for dynamic fracture tests with SHPB. Among them, the NSCB method is the most widely used because of its simple geometry, ease of manufacturing and convenient testing procedures [23,32]. Hence, the NSCB method is utilized to investigate the fracture behavior of KM sandstone samples in this study.

To minimize the variations of properties between samples and the scatters of test results, all samples were obtained from one sandstone block. According to the ISRM specification [23], cylindrical rock cores 49 mm in diameter were first drilled and then cut into discs 20 mm in thickness. Each discs was sawn along the diameter into two halves, i.e., semi-circular disc. After that, a notch with a length of 5 mm was carved onto the semi-circular disc from the center of the flat end of the semi-circular disc perpendicular to the diametrical cut. Finally, the notch tip was sharpened into a pre-fabricated crack tip by using a diamond wire with a diameter of 0.1 mm [33,34]. The schematic of the NSCB sample is shown in Fig. 2.

In our prior studies [2,3,35], the required time duration for the KM sandstone sample reaching water saturation by free soaking and then returning to dry state by air-drying are 24 h (1 day) and 144 h (6 days), respectively. Here, one artificial wetting-drying treatment in laboratory consists two processes, as illustrated in Fig. 3:

(1) saturation process: the dry samples were placed inside a tank filled with de-ionized water at 25 °C for 24 h for free-soaking;
(2) drying process: after saturation process, the saturated samples were taken out from the water tank and placed in a room with a constant temperature of 25 °C over 6 days for air-drying.

In this study, five experimental groups of sandstone samples respectively experiencing 10, 20, 30, 40 and 50 wetting-drying cycles were designed. One group of samples free from cyclic wetting-drying treatments (considered as zero cycle) were used for the contrastive analysis.

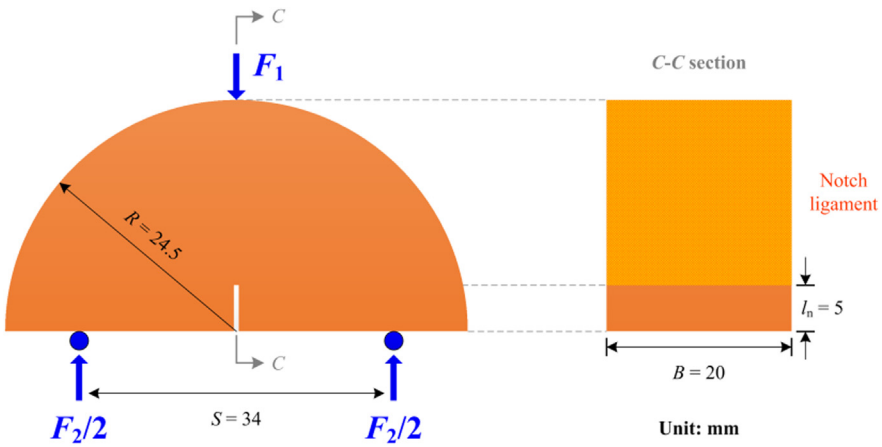


Fig. 2. Schematic of the NSCB sample.

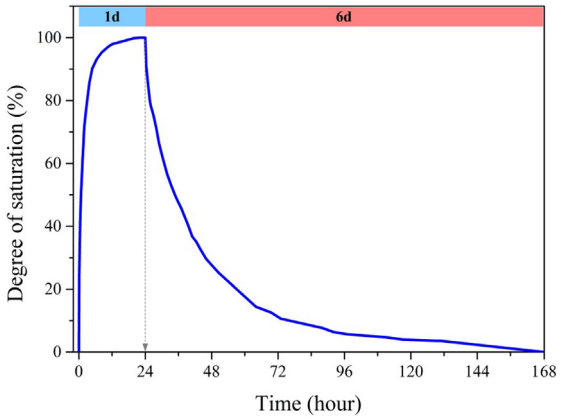


Fig. 3. Schematic of change in degree of saturation in the sample during one wetting-drying cycle.

2.3. Test setup and data acquisition

2.3.1. Quasi-static fracture tests

Conventional NSCB tests were conducted to obtain the quasi-static fracture toughness of KM sandstone samples after different wetting-drying cycles for comparison with the dynamic tests. All tests were performed on a servo-controlled material testing machine (MTS Insight 30) at the Advanced Research Center at Central South University. In tests, the displacement rate was kept constant at 0.05 mm/min. The axial load and displacement were automatically recorded by using a specific data acquisition frequency of 100 Hz.

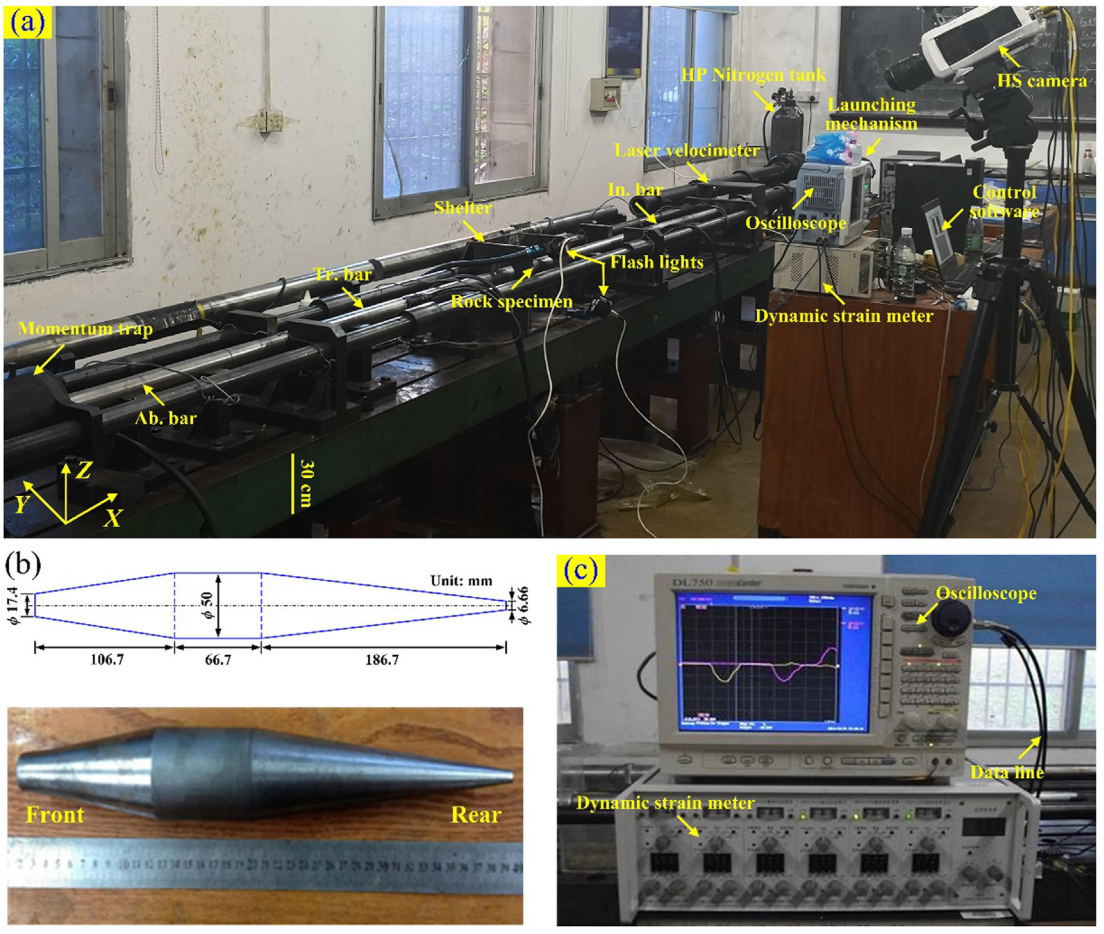


Fig. 4. Photograph of (a) the SHPB apparatus, (b) cone-shaped striker and (c) data acquisition system.

According to the ISRM standard, the maximum force (F_{max}) during loading process is used to calculate the quasi-static fracture toughness (K_{IC}^{st}) as [32]:

$$K_{IC}^{st} = Y' \frac{F_{max}\sqrt{\pi l_n}}{2RB} \tag{1}$$

where l_n is the pre-fabricated notch length; R is the radius of the sample; B is the thickness of the sample and Y' is a non-dimensional stress intensity factor related to sample geometry [32]:

$$Y' = -1.297 + 9.516\alpha - (0.47 + 16.457\alpha)\beta + (1.071 + 34.401\alpha)\beta^2 \tag{2}$$

where $\alpha = S/2R$, $\beta = l_n/R$ and S is the distance between the two supporting pins (see Fig. 2).

2.3.2. Split Hopkinson pressure bar

Dynamic NSCB tests were performed on an improved split Hopkinson pressure bar (SHPB) apparatus custom built at the Shock and Impact laboratory in the School of Resources and Safety Engineering of the Central South University. As displayed in Fig. 4a, the SHPB apparatus consists of three center-aligned steel bars, i.e., an incident bar (2 m in length), a transmitted bar (1.5 m in length) and an absorption bar (0.5 m in length), all of which are made of 40Cr alloy and cylinders 50 mm in diameter. The NSCB samples were sandwiched between the incident and transmitted bars. Sufficient grease was lubricated at bar/sample contacts to minimize the frication effect [36].

The cone-shaped striker (Fig. 4b) invented by Li et al. [37] was used in this study. It can produce a half-sine stress wave to eliminate high frequency oscillations, minimize the dispersion effect and ensure stress equilibrium. During tests, the striker which is launched by the gas gun impacts the front end of the incident bar and generates a half-sine stress wave (also called incident wave). The incident wave travels along the length of the incident bar towards the sample. Once the wave reaches the incident bar/sample interface, part of it is reflected back to the incident bar (called reflected wave), while the remaining part passes through the sample into the transmitted bar (called transmitted wave). The three stress waves were monitored by two pairs of dynamic strain gauges mounted at the middle of the incident and transmitted bars, respectively and recorded by a digital oscilloscope after signal transformation by a dynamic strain meter (Fig. 4c). Further details on the SHPB apparatus can be referred to previous publications [27,34,38–40].

According to 1D stress wave theory, the dynamic forces on the both ends of the NSCB sample can be calculated as [34]:

$$\begin{cases} F_1(t) = A_b E_b [\varepsilon_I(t) + \varepsilon_R(t)] \\ F_2(t) = A_b E_b \varepsilon_T(t) \end{cases} \tag{3}$$

where F_1 and F_2 are the forces at the incident bar/sample and transmitted bar/sample interfaces, respectively; A_b is the cross-sectional area of bars; E_b is the elastic modulus of bars, 233 GPa is used in this study; ε_I, ε_R and ε_T are the incident, reflected and transmitted signals, respectively.

If the stress equilibrium is reached, the dynamic stress intensity factor (SIF) at the notch tip of the NSCB sample can be determined as the quasi-static form [23]:

$$K_I^{dy}(t) = Y' \frac{\bar{F}(t)\sqrt{\pi l_n}}{2RB} \tag{4}$$

where K_I^{dy} is dynamic SIF; $\bar{F} = (F_1 + F_2)/2$ is the average dynamic force on both sides of sample. Similarly, the peak of dynamic stress intensity factor is defined as the dynamic fracture toughness (K_{IC}^{dy}).

Neglecting the energy consumption at the bar/sample interfaces, the energy dissipation of rock sample (W_s) can be expressed as

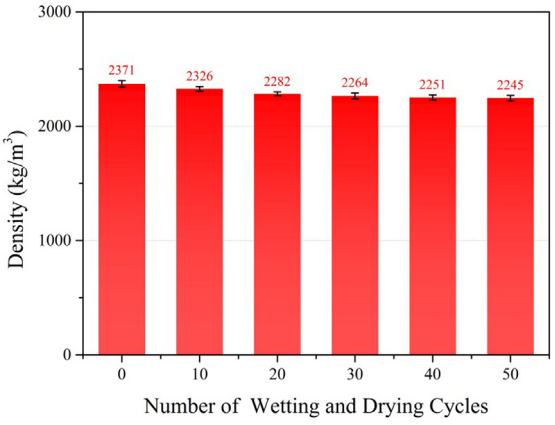


Fig. 5. The variation in average density after wetting-drying cycles.

[41–44]:

$$W_s = W_I - W_R - W_T \tag{5}$$

where W_I, W_R and W_T are the elastic strain energy in bars induced by incident, reflected and transmitted waves during the stress wave duration, their expressions are as following [43–45]:

$$\begin{cases} W_I = A_b C_b E_b \int \varepsilon_I(t)^2 dt \\ W_R = A_b C_b E_b \int \varepsilon_R(t)^2 dt \\ W_T = A_b C_b E_b \int \varepsilon_T(t)^2 dt \end{cases} \tag{6}$$

where C_b is the P-wave velocity of bars, 5450 m/s is used.

3. Test results

3.1. Cumulative damage induced by wetting-drying cycles

3.1.1. Density

Fig. 5 presents the average density of the KM sandstone subjected to cyclic wetting-drying. It is clearly observed that the density of sandstone decreases with the increase of cycle number. Compared with the untreated sample, the density is reduced by 2.16%, 3.86%, 4.38%, 4.80% and 5.38% after 10, 20, 30, 40 and 50 wetting-drying cycles, respectively.

3.1.2. Porosity

The porosity of samples was measured by using the nuclear magnetic resonance (NMR) technique (MiniMR-60). Fig. 6 shows the porosity of KM sandstone after different number of wetting-drying cycles. The initial porosity (before cyclic wetting-drying treatments) of KM sandstone is about 8.87%. With the increase of cycle number, the porosity gradually increases. When 50 cycle is reached, the average value of porosity hits 10.14% (showing an increase by 14.3%). This indicates that new microcracks generated and irreversible damage accumulated in rock samples during cyclic wetting-drying.

3.1.3. P-wave velocity

The P-wave velocity provides a crucial indicator for rock properties. We also measured the P-wave velocity of rock samples after every 10 wetting-drying cycles, as plotted in Fig. 7. In contrast to the porosity, the average P-wave velocity of KM sandstone decreases with the increasing number of wetting-drying cycles. For initial samples, the average P-wave velocity is about 2813 m/s. When the cyclic number is ten, the P-wave velocity of KM sandstone significantly drops by 13.0% to a value of 2447 m/s. It is observed that after 10 cycles, the drop of P-wave velocity seems to be more slight compared to the first ten cycles.

Fig. 8 shows the P-wave velocity decreases with the increase of porosity. The relationship between the P-wave velocity (C_p) and porosity (ϕ) follows an exponential rule:

$$C_p(\phi) = 2235 + 3.597e^{-2.54\phi} \ (R^2 = 0.999) \tag{7}$$

where R^2 is the correlation coefficient.

The ultrasonic wave will refract at inclined defects. A large amount of new-formed microcracks induced by cyclic wetting-drying increases the travelling path of the ultrasonic wave to some extent, which would lead to the decrease the P-wave velocity of sample.

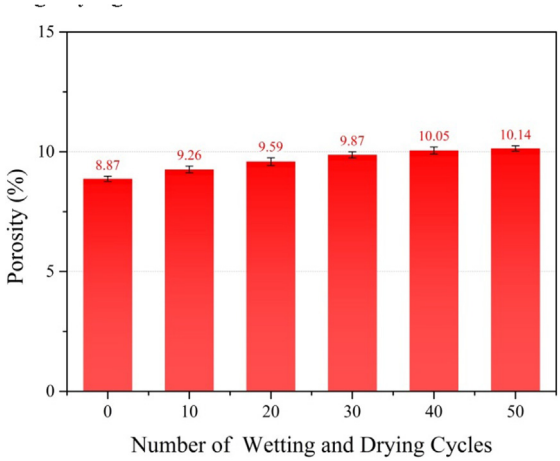


Fig. 6. The variation in average porosity after wetting-drying cycles.

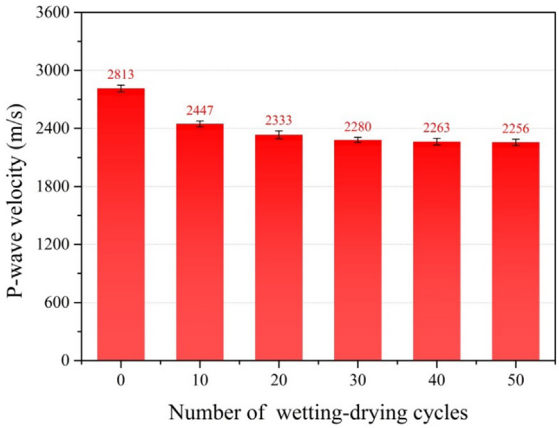


Fig. 7. The variation in average P-wave velocity after wetting-drying cycles.

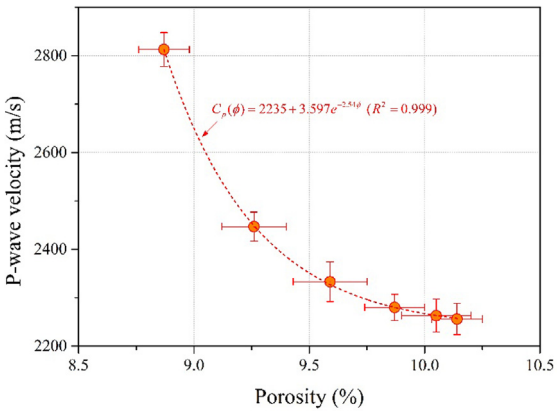


Fig. 8. P-wave velocity versus porosity.

3.1.4. Microstructural observations

The microstructural features of sandstone samples after different cyclic wetting-drying treatments were obtained by using scanning electron microscopy (SEM) to reveal the cumulative damage. Fig. 9 displays some representative SEM images with the magnification of 3000. For the untreated rock sample (see Fig. 9a), the large quartz grains are well cemented and connect tightly with each other via face-to-face contact. There are no visible cracks or fractures in the rock samples without cyclic wetting-drying except for little initial fissures. From Fig. 9b, when the rock sample is subjected to 10 wetting-drying cycles, a new-formed intergranular crack can be observed and the finer granules become looser. After 30 wetting-drying cycles, more intergranular cracks generate. The length and density of microcracks also increase. The amount of cementation between grains apparently decreases (Fig. 9c). As shown in Fig. 9d, for the sample with 50 cyclic wetting-drying, the pore spaces between grains significantly increase. The maximum width of intergranular cracks even increases to more than 20 μm. The adjacent quartz grains are cemented by a very piece of calcite, i.e., the junction between grains changes to point-contact. These microstructural observations imply that wetting-drying cycles can weaken the connections between mineral grains.

3.2. Fracture properties of sandstone after wetting-drying cycles

3.2.1. Quasi-static fracture behavior

Fig. 10 shows the load-displacement curves of sandstone samples after wetting-drying cycles. It is evident that the deformation processes of all samples experience four phases: (1) crack closure phase, where pre-existing defects inside the rock sample (including pores, cracks and fissures) are squeezed. This results in that the initial stage of the load-displacement curve is concave; (2) elastic deformation phase, where the deformation increases linearly as the external load rises; (3) crack growth phase. After nearly 80–90% of peak load, the applied load is high enough to induce the growth and nucleation of micro-cracks. The curves thus become non-linear; (4) post-peak phase, where the loads of samples suddenly drop after peak load. This means that the NSCB sample totally loses its bearing capacity. It is observed that, the wetting-drying cycles lead to significant impacts on the first two phases. As the number of wetting-drying cycles increases, the crack closure phase is more obvious. Also, the slope of load-displacement curve in elastic regime decreases and indicates the loss of stiffness. These phenomena are mainly caused by micro-cracks growth and expansion in rock

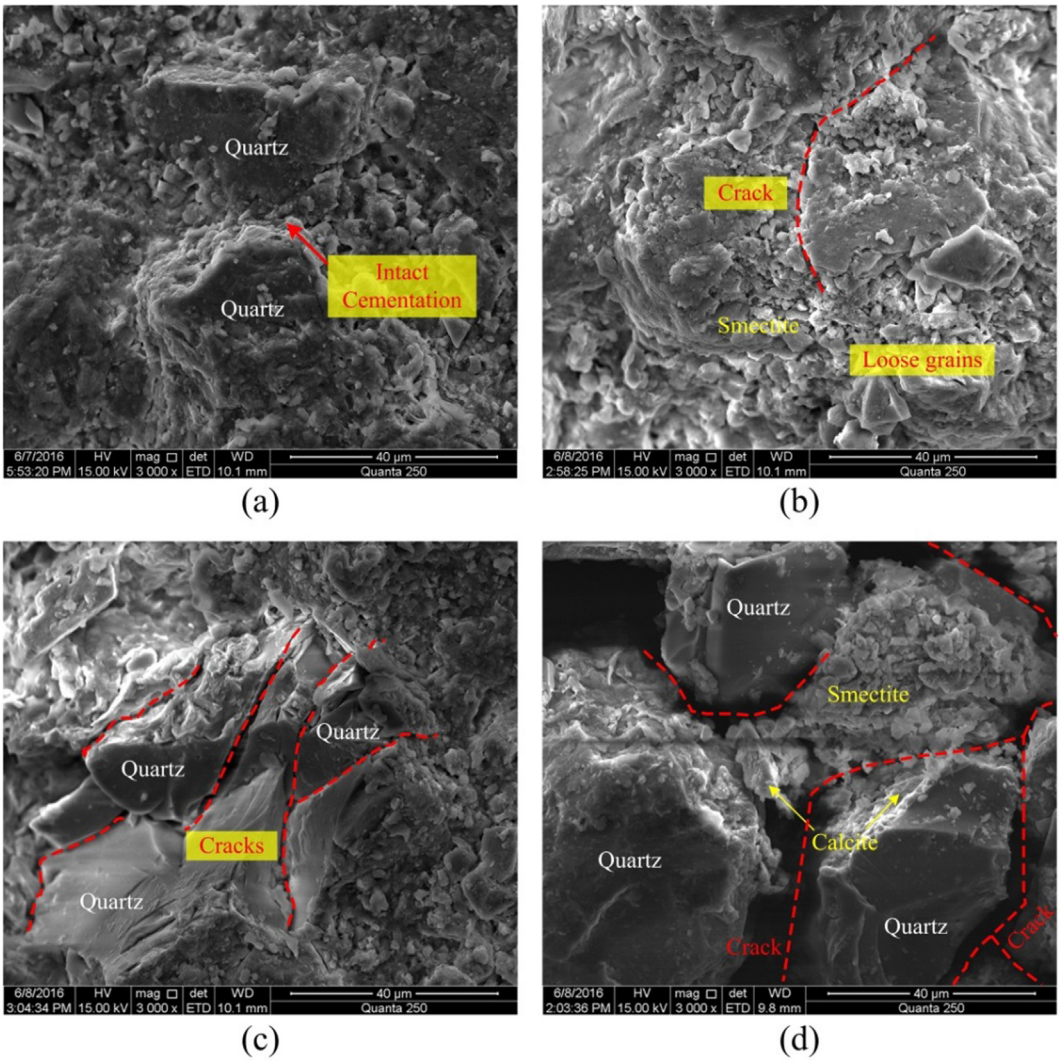


Fig. 9. SEM images of KM sandstone subjected to (a) 0, (b) 10, (c) 30 and (d) 50 wetting-drying cycles.

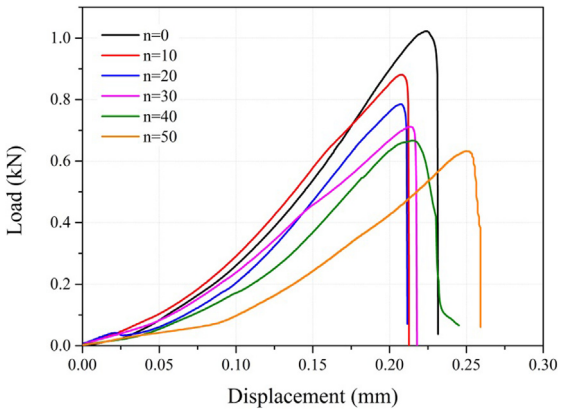


Fig. 10. Load-displacement curves of sandstone samples after *n* wetting-drying cycles.

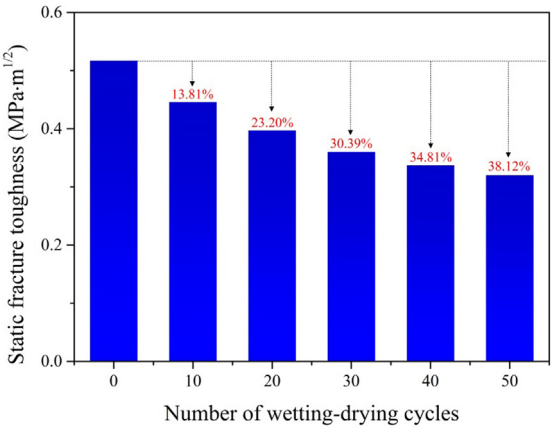


Fig. 11. Quasi-static fracture toughness of sandstone samples after wetting-drying cycles.

samples during cyclic wetting-drying treatments (see Fig. 9).

According to Eqs. (1) and (2), the quasi-static fracture toughness of sample was calculated. As shown in Fig. 11, the quasi-static fracture toughness of KM sandstone gradually decreases with the increasing number of wetting-drying cycles. The quasi-static fracture toughness of untreated sample is about 0.52 MPa·m$^{1/2}$. When the cycle number are 10, 20, 30, 40 and 50, it is reduced by 13.81%, 23.20%, 30.39%, 34.81% and 38.72%, respectively. The pattern of fracture toughness evolution is quite similar to that of P-wave velocity (see Fig. 7), i.e., the decreasing rate is highest in the first ten cycles but attenuates as the cycle number increases. During wetting-drying cycles, water-rock interaction occurs and weakens the cementation between grains, further resulting in micro-cracks growth and expansion. At the beginning of cyclic wetting-drying, the water-rock interaction is very strong so that the fracture toughness fast decreases. However, with the increase of cycle number, the water-rock interaction gradually weakens, and the rate of fracture toughness decrease tends to slow down.

3.2.2. *Dynamic fracture toughness*

Fig. 12 illustrates the variation of the dynamic fracture toughness of sandstone samples subjected to different wetting-drying cycles versus loading rate. Similar with quasi-static fracture toughness, for a given loading rate, the initial sandstone sample has the maximum value of dynamic fracture toughness. The fracture toughness of samples also decreases with the increase of the number of wetting-drying cycles. This indicates that the cyclic wetting-drying treatments can deteriorate the mechanical properties of sandstone. Moreover, the dynamic fracture toughness of sandstone is dependent on loading rate regardless of cycle number. In other word, it increases with the increase of loading rate. This may be attributed to the structural inertia effect and the viscous behavior of the bulk material between cracks [29,34,46]. Generally, the dynamic fracture toughness of rock against loading rate follows the power law as [34,47–49]:

$$K_{ICn}^{dy}(\dot{K}_I) = K_{ICn}^{st}\cdot\dot{K}_I{}^{a} \tag{8}$$

where K_{ICn}^{dy} and K_{ICn}^{st} are the dynamic and quasi-static fracture toughness of samples after n wetting-drying cycles; $\dot{K}_I$ is the loading

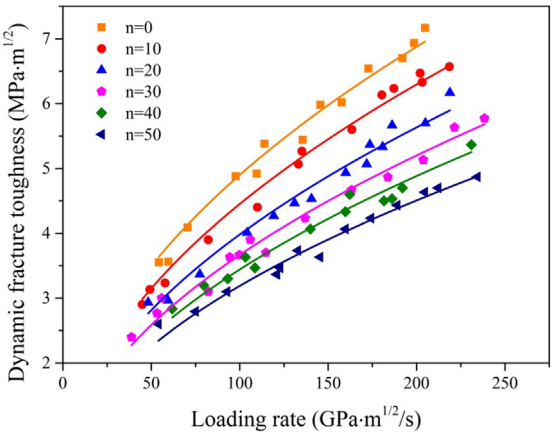


Fig. 12. Dynamic fracture toughness of KM sandstone after different wetting-drying cycles versus loading rate (solid symbols indicate the tested data and solid lines denote the fitting curves).

rate; a is the fitting constant. The fitting curves obtained by the model are plotted in Fig. 12, showing that the power model characterizes the rate dependence of dynamic fracture toughness of rock well. The fitting formulae are listed as following:

$$\begin{cases} K_{IC0}^{dy}(\dot{K}_I) = 0.517\dot{K}_I^{0.488} \ (R^2 = 0.984) \\ K_{IC10}^{dy}(\dot{K}_I) = 0.446\dot{K}_I^{0.499} \ (R^2 = 0.991) \\ K_{IC20}^{dy}(\dot{K}_I) = 0.397\dot{K}_I^{0.501} \ (R^2 = 0.979) \\ K_{IC30}^{dy}(\dot{K}_I) = 0.360\dot{K}_I^{0.504} \ (R^2 = 0.980) \\ K_{IC40}^{dy}(\dot{K}_I) = 0.337\dot{K}_I^{0.505} \ (R^2 = 0.970) \\ K_{IC50}^{dy}(\dot{K}_I) = 0.320\dot{K}_I^{0.499} \ (R^2 = 0.975) \end{cases} \tag{9}$$

3.2.3. Energy dissipation

Fig. 13 shows the energy absorbed by samples during dynamic tests according to Eqs. (5) and (6). It is observed that, the energy dissipation shows a clear increasing manner with respect to the loading rate under a same cycle number. This could be due to the fact that with the increase of loading rate, more energy is consumed for the increase of new-formed micro and macro fractures, the increase of fracture energy per area and the higher velocity of flying fragments (kinetic energy) [34,42,50]. It can also be seen from Fig. 13 that, under a given loading rate, the energy dissipation gradually decreases as the number of cyclic wetting-drying increases. This is likely attributed to the higher crack density and weaker grain cementation caused by wetting-drying cycles.

4. Discussion

Numerous studies suggest that water can significantly weaken the rock strength and fracture toughness through series of physical, chemical and mechanical water-rock interactions [33,34,51–54]. The majority of water-weakening effects are reversible when the saturated rock sample gets dry state again [35], but some are not, such as calcite solution and swelling clay induced microcracking. The cyclic wetting-drying can further aggravate the irreversible effects and result in cumulative damage, which can be revealed by the microscopic morphological characteristics of rock sample (see Fig. 13). The underlying mechanisms will be interpreted as following:

(1) calcite dissolution: in sample preparation, rock samples are saturated by de-ionized water, whose pH value is close to 7.0. Calcite is generally regarded as insoluble in de-ionized water under vacuum condition. However, the cyclic wetting-drying treatments are performed under atmospheric condition in this study. As suggested by Haynes [55], the solubility of calcite under atmospheric condition is 2.4 times greater than that under vacuum condition. The air contains gaseous carbon dioxide (CO_2). Water can absorb CO_2 from the air and generate hydrogen ion (H^+). This will increase the solubility of calcite significantly. The dissolution process of calcite involves the following chain of reactions [56]:

$$H_2O + CO_2(g) \rightleftharpoons H^+ + HCO_3^- \tag{10}$$

$$CaCO_3 + H^+ \rightleftharpoons Ca^{2+} + HCO_3^- \tag{11}$$

It is noted that the above-mentioned reactions occur placidly and slowly. In this study, one complete wetting-drying process lasts over seven days. The duration of wetting-drying cycles is sufficient for the calcite dissolution. Therefore, the calcite cementation between grains gradually weakens as the increasing cycle number.

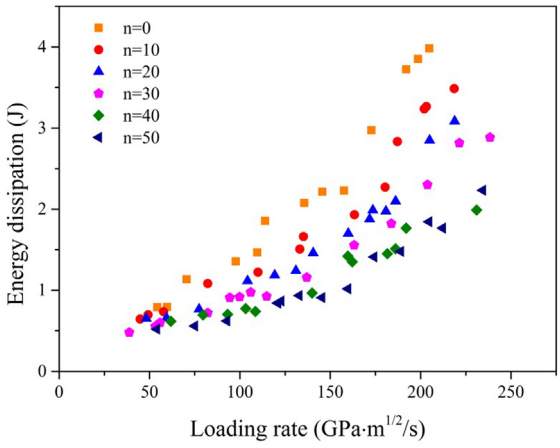


Fig. 13. Energy dissipation of sample after different wetting-drying cycles versus loading rate.

(2) intergranular cracking: as a typical expandable clay mineral, smectite is abundant in the KM sandstone. The creation and widening of intergranular cracks are highly relevant to the swelling of smectite Assuming that the intergranular spaces are filled with smectite such that the micromechanical model can be simplified as the rock skeleton containing an expandable inclusion (see Fig. 14a). When the rock sample is submerged in water, the smectite will absorb water molecules and dilate [57,58]. Due to the volume of smectite restricted by pore size, the swelling pressure (σ_{sp}) develops and exerts on the pore wall. The swelling pressure is considered as uniform, so it is like pore-water pressure. As shown in Fig. 14b, the approximate solution of the SIF at crack tip induced by σ_{sp} can be expressed as [59]:

$$K_{sp} = \sigma_{sp}\sqrt{r}\left(\frac{1.3L}{1+(1+L)^{3/2}} + \frac{7.8\sin(L/2)}{2(1+L)^{3/2}-1.7}\right) \tag{12}$$

where K_{sp} is the SIF induced by σ_{sp}; r is the pore radius; $L = l/r$ is the normalized crack length; l is the length of subcritical crack. When K_{sp} is higher than the SIF threshold for subcritical cracking (K_{th}) but lower than the quasi-static fracture toughness (K_{IC}^{st}), the subcritical cracks will emanate from the pores [59].

The swelling pressure can be determined as[60]:

$$\sigma_{sp} = \frac{E_c \varepsilon_c}{1-\nu_c} \tag{13}$$

where E_c, ε_c and ν_c are the elastic modulus, swelling strain and poisson ratio of smectite, respectively.

To approximately obtain the SIF induced by the swelling pressure under water saturation condition, the following parameters were adopted as: $\varepsilon_c = 0.8$ [61] and $\nu_c = 0.5$. E_c ranges 5 MPa to 15 MPa [62]. According to Eq.(13), the swelling pressure induced by water saturation is between 8 MPa and 24 MPa. In addition, $L = 0.2$ [63] and r ranges from 5 μm to 100 μm. The SIF can thus be calculated according to Eq.(12) as shown in Fig. 15. The SIF is positively dependent with both swelling pressure and pore radius. The maximum value of SIF is lower than the quasi-static fracture toughness (K_{IC}^{st}). This means that the sandstone sample will not generate macro fractures and disintegrate during cyclic wetting-drying. As for the SIF threshold for subcritical cracking (K_{th}), there is no consensus on its magnitude [64]. It is most commonly held that K_{th} of rock materials is on the order of 10–20% of its K_{IC}^{st}[65,66]. Hence, for the KM sandstone, K_{th} ranges from 0.052 to 0.104 MPa·m$^{1/2}$ under dry state. From Fig. 15, it is also evident that the major portion of SIF curves is above the K_{th} limits. This means that subcritical cracking will occur in the pores whose SIF exceeding K_{th} after water saturation so as to increase the crack density in rock sample. In fact, the presence of water prominently weakens the K_{th} of rock, which will further facilitate the initiation and propagation of microcracks induced by clay swelling [66,67].

According to Eq.(13), the swelling pressure is proportion to the swelling strain. During saturation process, the swelling strain of smectite increases with the increase in the degree of saturation, so the swelling pressure reaches the maximum value when the rock sample is completely saturated. However, during drying process, the swelling strain of smectite gradually recovers with the evaporation of water. The swelling pressure decreases accordingly. Actually, the subcritical cracks are subjected to repeated tensile loads during cyclic wetting-drying treatments, as shown in Fig. 16. The cyclic tension will lead to fatigue damage on rock pore structure and decrease the fracture threshold [68–73]. In addition, the non-uniform and non-synchronous swelling and shrinking of clay minerals in rock sample further aggravate internal microcracks growth and expansion [3]. It should be noted that the intergranular cracking process depends on the content of expansive clay mineral. Prior researches suggest that high clay-bearing rocks are prone to be deteriorated by wetting-drying cycles [12,14,21,22]. However, for rocks with very low clay proportion, the mechanical properties are not sensitive to cyclic wetting-drying [11,16,18]. This is highly consistent with our hypothesis.

The dissolution of calcite cementation and the increase of crack density will decrease the fracture toughness of rock sample from the following two aspects:

(1) According to Griffith fracture theory, the fracture toughness of rock sample is theoretically proportional to the elastic modulus

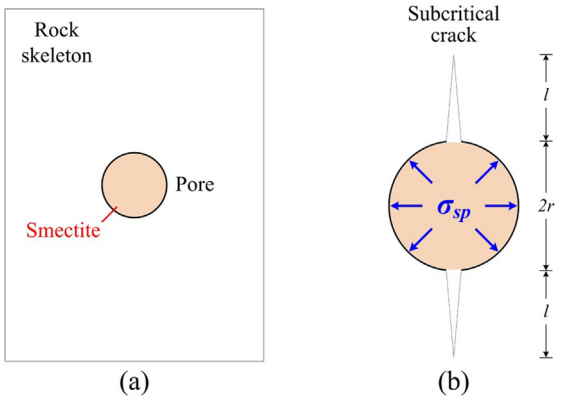


Fig. 14. (a) micromechanical model for intergranular cracking and (b) subcritical cracks induced by clay swelling.

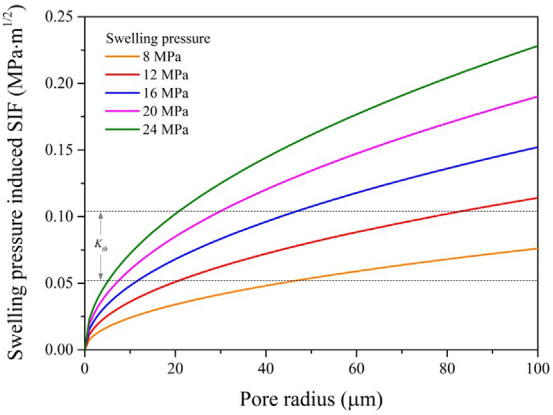


Fig. 15. Stress intensity factor induced by swelling pressure under water-saturated condition (the black dash lines denote the K_{th} range).

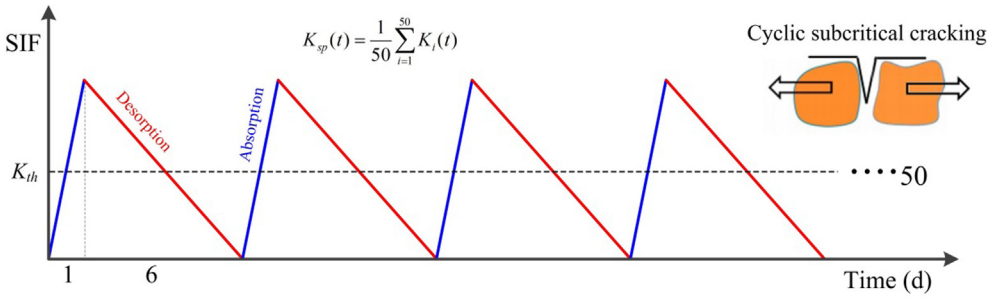


Fig. 16. Simplified schematic of clay swelling induced SIF evolution during cyclic wetting-drying treatments.

(E) and specific surface energy (γ) [74]:

$$K_{IC} \propto \sqrt{E\gamma} \tag{14}$$

It is speculated that the wetting-drying cycles affect the elastic modulus of sample. Herein, the elastic modulus of rock sample after wetting-drying cycles is not experimentally measured but can be simply estimated by the density (ρ) and P-wave velocity (C_p) as:

$$E = \rho C_p^{\,2} \tag{15}$$

From Fig. 17, the elastic modulus shows a clear decreasing trend against the number of wetting-drying cycles. This agrees with the test results by previous studies [2,9,12,14]. The specific surface energy (γ), the other parameter in Eq. (14), also decreases with cycle number (see Fig. 13). These give rise to the reduction in the fracture toughness of the sandstone sample.

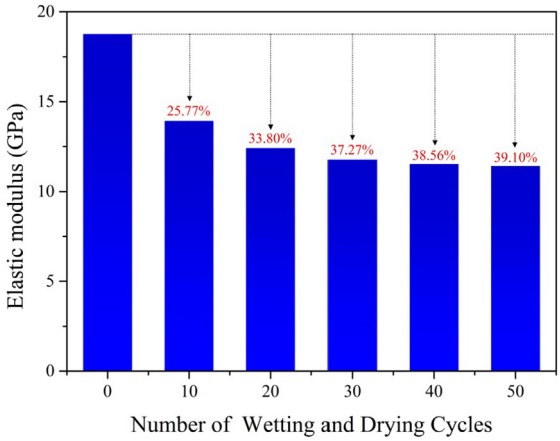


Fig. 17. Estimated elastic modulus of rock sample after wetting-drying cycles.

(2) At the microscale, the NSCB test can be regarded as the problem of a main crack interacting with a microcrack network [75]. The SIF at the tip of the main crack dominates the fracture of rock sample. Some rigorous theoretical studies suggested that the SIF at the main crack tip exponentially increases as the microcrack spacing decreases [76–79]. In other words, a more dense microcrack network in the near field can induce larger SIF amplification at the crack tip, so that the fracture initiation and propagation occur under a lower external load. The SEM images (Fig. 9) indicate that as cycle number increases, the space between each microcracks reduces. This causes the apparent reduction of fracture toughness.

For rock engineering constructions, especially in pluvial regions, rock structures usually suffer the water erosion and cyclic wetting-drying deterioration. Several potential protection methods are suggested to avoid or reduce the negative effects of water: (1) the water-proof facilities should be installed at critical structures; (2) some specific rock binders can be utilized to repair the visible and macroscopic fractures on the surface and inside rocks; (3) the rock materials with high proportion of expandable and soluble minerals cannot be adopted for the critical structures.

5. Conclusions

To investigate the effect of repeated wetting-drying on the fracture behavior of sandstone, quasi-static and dynamic NSCB tests were performed on the KM sandstone samples after every ten artificial wetting-drying cycles. The fracture toughness and energy dissipation were obtained and compared. The density, porosity and P-wave velocity of sample were determined. The cumulative damage in rock sample induced by cyclic wetting-drying was revealed by SEM images. The deterioration mechanisms were also discussed. The following conclusions can be drawn:

(1) Cyclic wetting-drying remarkably affects the physical properties of KM sandstone. As the number of wetting-drying cycles increases, the density and P-wave velocity of sandstone gradually decrease and the porosity increases. The variation of each physical parameter is the most significant in the first ten cycles, afterwards the changing rate becomes gentle.
(2) The SEM images of rock samples after different cycle number suggest that the density and interconnectivity of the internal microcracks increase with wetting-drying cycles. The tight surface-contact between grains is gradually transformed to the weak point-contact.
(3) Under the same loading rate, the fracture toughness and energy dissipation of rock decreases as the increasing cycle number. At a given cycle number, the fracture toughness and energy dissipation increase with the increase of loading rate. The primary mechanisms for the rock deterioration after wetting-drying cycles are the dissolution of calcite and the intergranular cracking induced by clay swelling.

Acknowledgments

The work is supported by financial grants from the National Basic Research Program of China (2015CB060200), the National Natural Science Foundation of China (41772313), the Natural Science Foundation of Hunan (2015JJ4067). The authors are very grateful to the financial contribution and convey their appreciation for supporting this basic research.

Declaration of Competing Interest

The authors declare that they have no conflict of interest.

References

[1] Gökceoglu C, Ulusay R, Sönmez H. Factors affecting the durability of selected weak and clay-bearing rocks from Turkey, with particular emphasis on the influence of the number of drying and wetting cycles. Eng Geol 2000;57:215–37. https://doi.org/10.1016/S0013-7952(00)00031-4.
[2] Zhou Z, Cai X, Chen L, Cao W, Zhao Y, Xiong C. Influence of cyclic wetting and drying on physical and dynamic compressive properties of sandstone. Eng Geol 2017:220. https://doi.org/10.1016/j.enggeo.2017.01.017.
[3] Zhou Z, Cai X, Ma D, Chen L, Wang S, Tan L. Dynamic tensile properties of sandstone subjected to wetting and drying cycles. Constr Build Mater 2018;182:215–32. https://doi.org/10.1016/j.conbuildmat.2018.06.056.
[4] Ma D, Duan H, Li X, Li Z, Zhou Z, Li T. Effects of seepage-induced erosion on nonlinear hydraulic properties of broken red sandstones. Tunn Undergr Sp Technol 2019. https://doi.org/10.1016/j.tust.2019.102993.
[5] Dang W, Wu W, Konietzky H, Qian J. Effect of shear-induced aperture evolution on fluid flow in rock fractures. Comput Geotech 2019;114:103152https://doi.org/10.1016/j.compgeo.2019.103152.
[6] Momeni A, Hashemi SS, Khanlari GR, Heidari M. The effect of weathering on durability and deformability properties of granitoid rocks. Bull Eng Geol Environ 2017;76:1037–49. https://doi.org/10.1007/s10064-016-0999-7.
[7] Hale PA, Shakoor A. A laboratory investigation of the effects of cyclic heating and cooling, wetting and drying, and freezing and thawing on the compressive strength of selected sandstones. Environ Eng Geosci 2003;9:117–30. https://doi.org/10.2113/9.2.117.
[8] Yao HY, Zhang ZH, Zhu ZH. Uniaxial mechanical properties of sandstone under cyclic of drying and wetting. Adv Mater Res 2011;243–249:2310–3. https://doi.org/10.4028/www.scientific.net/AMR.243-249.2310.
[9] Yao W, Li C, Zhan H, Zhou JQ, Criss RE, Xiong S, et al. Multiscale study of physical and mechanical properties of sandstone in three gorges reservoir region subjected to cyclic wetting-drying of Yangtze river water. Rock Mech Rock Eng 2020. https://doi.org/10.1007/s00603-019-02037-7.
[10] Qin Z, Chen X, Fu H. Damage features of altered rock subjected to drying-wetting cycles. Adv Civ Eng 2018. https://doi.org/10.1155/2018/5170832.
[11] Zhao Z, Yang J, Zhang D, Peng H. Effects of wetting and cyclic wetting-drying on tensile strength of sandstone with a low clay mineral content. Rock Mech Rock Eng 2017;50:485–91. https://doi.org/10.1007/s00603-016-1087-9.
[12] Liu X, Wang Z, Fu Y, Yuan W, Miao L. Macro / microtesting and damage and degradation of sandstones under dry-wet cycles. Adv Mater Sci Eng 2016. https://

doi.org/10.1155/2016/7013032.
[13] Zhang BY, Zhang JH, Sun GL. Deformation and shear strength of rockfill materials composed of soft siltstones subjected to stress, cyclical drying/wetting and temperature variations. Eng Geol 2015;190:87–97. https://doi.org/10.1016/j.enggeo.2015.03.006.
[14] Huang S, Wang J, Qiu Z, Kang K. Effects of cyclic wetting-drying conditions on elastic modulus and compressive strength of sandstone and mudstone. Processes 2018. https://doi.org/10.3390/pr6120234.
[15] Pardini G, Guidi GV, Pini R, Regüés D, Gallart F. Structure and porosity of smectitic mudrocks as affected by experimental wetting-drying cycles and freezing-thawing cycles. Catena 1996;27:149–65. https://doi.org/10.1016/0341-8162(96)00024-0.
[16] Lin ML, Jeng FS, Tsai LS, Huang TH. Wetting weakening of tertiary sandstones - Microscopic mechanism. Environ Geol 2005;48:265–75. https://doi.org/10.1007/s00254-005-1318-y.
[17] Loubser PD, Sumner M. Experimental sandstone weathering using different wetting and drying moisture amplitudes. Earth Surf Process Landforms 2008;33:985–90. 10.1002/esp.
[18] Özbek A. Investigation of the effects of wetting–drying and freezing–thawing cycles on some physical and mechanical properties of selected ignimbrites. Bull Eng Geol Environ 2014;73:595–609. https://doi.org/10.1007/s10064-013-0519-y.
[19] Zhang Z, Jiang Q, Zhou C, Liu X. Strength and failure characteristics of Jurassic Red-Bed sandstone under cyclic wetting-drying conditions. Geophys J Int 2014;198:1034–44. https://doi.org/10.1093/gji/ggu181.
[20] Liu X, Jin M, Li D, Zhang L. Strength deterioration of a Shaly sandstone under dry–wet cycles: a case study from the Three Gorges Reservoir in China. Bull Eng Geol Environ 2017:1–15. https://doi.org/10.1007/s10064-017-1107-3.
[21] Khanlari G, Abdilor Y. Influence of wet–dry, freeze–thaw, and heat–cool cycles on the physical and mechanical properties of Upper Red sandstones in central Iran. Bull Eng Geol Environ 2015;74:1287–300. https://doi.org/10.1007/s10064-014-0691-8.
[22] Hua W, Dong S, Li Y, Xu J, Wang Q. The influence of cyclic wetting and drying on the fracture toughness of sandstone. Int J Rock Mech Min Sci 2015;78:331–5. https://doi.org/10.1016/j.ijrmms.2015.06.010.
[23] Zhou YX, Xia K, Li XB, Li HB, Ma GW, Zhao J, et al. Suggested methods for determining the dynamic strength parameters and mode-I fracture toughness of rock materials. Int J Rock Mech Min Sci 2012;49:105–12. https://doi.org/10.1016/j.ijrmms.2011.10.004.
[24] Hua W, Dong S, Li Y, Wang Q. Effect of cyclic wetting and drying on the pure mode II fracture toughness of sandstone. Eng Fract Mech 2016;153:143–50. https://doi.org/10.1016/j.engfracmech.2015.11.020.
[25] Hua W, Dong S, Peng F, Li K, Wang Q. Experimental investigation on the effect of wetting-drying cycles on mixed mode fracture toughness of sandstone. Int J Rock Mech Min Sci 2017;93:242–9. https://doi.org/10.1016/j.ijrmms.2017.01.017.
[26] Cai X, Zhou Z, Du X. Water-induced variations in dynamic behavior and failure characteristics of sandstone subjected to simulated geo-stress. Int J Rock Mech Min Sci 2020;130:104339https://doi.org/10.1016/j.ijrmms.2020.104339.
[27] Zhou Z, Cai X, Li X, Cao W, Du X. Dynamic response and energy evolution of sandstone under coupled static-dynamic compression: insights from experimental study into deep rock engineering applications. Rock Mech Rock Eng 2019. https://doi.org/10.1007/s00603-019-01980-9.
[28] Li X, Gong F, Tao M, Dong L, Du K, Ma C, et al. Failure mechanism and coupled static-dynamic loading theory in deep hard rock mining: A review. J Rock Mech Geotech Eng 2017;9:767–82. https://doi.org/10.1016/j.jrmge.2017.04.004.
[29] Zhang QB, Zhao J. A review of dynamic experimental techniques and mechanical behaviour of rock materials. Rock Mech Rock Eng 2014;47:1411–78. https://doi.org/10.1007/s00603-013-0463-y.
[30] Franklin JA, Sun ZQ, Atkinson BK, Meredith PC, Rummel F, Mueller W, et al. Suggested methods for determining the fracture toughness of rock. Int J Rock Mech Min Sci Geomech Abstr 1988;25:71–96. https://doi.org/10.1016/0148-9062(88)91871-2.
[31] Fowell RJ, Hudson JA, Xu C, Chen JF, Zhao X. Suggested method for determining mode I fracture toughness using Cracked Chevron Notched Brazilian Disc (CCNBD) specimens. Int J Rock Mech Min Sci 1995. https://doi.org/10.1016/0148-9062(94)00015-U.
[32] Kuruppu MD, Obara Y, Ayatollahi MR, Chong KP, Funatsu T. ISRM-suggested method for determining the mode i static fracture toughness using semi-circular bend specimen. Rock Mech Rock Eng 2014;47:267–74. https://doi.org/10.1007/s00603-013-0422-7.
[33] Zhou Z, Cai X, Ma D, Cao W, Chen L, Zhou J. Effects of water content on fracture and mechanical behavior of sandstone with a low clay mineral content. Eng Fract Mech 2018;193:47–65. https://doi.org/10.1016/j.engfracmech.2018.02.028.
[34] Zhou Z, Cai X, Ma D, Du X, Chen L, Wang H, et al. Water saturation effects on dynamic fracture behavior of sandstone. Int J Rock Mech Min Sci 2019;114:46–61. https://doi.org/10.1016/j.ijrmms.2018.12.014.
[35] Zhou Z, Cai X, Cao W, Li X, Xiong C. Influence of water content on mechanical properties of rock in both saturation and drying processes. Rock Mech Rock Eng 2016;49:3009–25. https://doi.org/10.1007/s00603-016-0987-z.
[36] Dai F, Huang S, Xia K, Tan Z. Some fundamental issues in dynamic compression and tension tests of rocks using split Hopkinson pressure bar. Rock Mech Rock Eng 2010;43:657–66. https://doi.org/10.1007/s00603-010-0091-8.
[37] Li XB, Lok TS, Zhao J. Dynamic characteristics of granite subjected to intermediate loading rate. Rock Mech Rock Eng 2005;38:21–39. https://doi.org/10.1007/s00603-004-0030-7.
[38] Weng L, Wu Z, Liu Q. Dynamic mechanical properties of dry and water-saturated siltstones under sub-zero temperatures. Rock Mech Rock Eng 2020. https://doi.org/10.1007/s00603-019-02039-5.
[39] Li XF, Li HB, Zhang QB, Jiang JL, Zhao J. Dynamic fragmentation of rock material: Characteristic size, fragment distribution and pulverization law. Eng Fract Mech 2018;199:739–59. https://doi.org/10.1016/j.engfracmech.2018.06.024.
[40] Han Z, Li D, Zhou T, Zhu Q, Ranjith PG. Experimental study of stress wave propagation and energy characteristics across rock specimens containing cemented mortar joint with various thicknesses. Int J Rock Mech Min Sci 2020;131:104352https://doi.org/10.1016/j.ijrmms.2020.104352.
[41] Chen R, Xia K, Dai F, Lu F, Luo SN. Determination of dynamic fracture parameters using a semi-circular bend technique in split Hopkinson pressure bar testing. Eng Fract Mech 2009;76:1268–76. https://doi.org/10.1016/j.engfracmech.2009.02.001.
[42] Zhang QB, Zhao J. Quasi-static and dynamic fracture behaviour of rock materials: Phenomena and mechanisms. Int J Fract 2014;189:1–32. https://doi.org/10.1007/s10704-014-9959-z.
[43] Song B, Chen W. Energy for specimen deformation in a split Hopkinson pressure bar experiment. Exp Mech 2006;46:407–10. https://doi.org/10.1007/s11340-006-6420-x.
[44] Weng L, Wu Z, Liu Q, Wang Z. Energy dissipation and dynamic fragmentation of dry and water-saturated siltstones under sub-zero temperatures. Eng Fract Mech 2019;220:106659https://doi.org/10.1016/j.engfracmech.2019.106659.
[45] Zhang Z, Kou S, Jiang L, Lindqvist P. Effects of loading rate on rock fracture: fracture characteristics and energy partitioning. Int J Rock Mech Min Sci 2000;37:745–62. https://doi.org/10.1016/S1365-1609(00)00008-3.
[46] Ožbolt J, Bošnjak J, Sola E. Dynamic fracture of concrete compact tension specimen: Experimental and numerical study. Int J Solids Struct 2013;50:4270–8. https://doi.org/10.1016/j.ijsolstr.2013.08.030.
[47] Lankford J. The role of tensile microfracture in the strain rate dependence of compressive strenght of fine-grained limestone-analogy with strong ceramics. Int J Rock Mech Min Sci 1981. https://doi.org/10.1016/0148-9062(81)90742-7.
[48] Ogata Y, Jung W, Kubota S, Wada Y. Effect of the strain rate and water saturation for the dynamic tensile strength of rocks. Mater Sci Forum 2004;465–466:361–6. https://doi.org/10.4028/www.scientific.net/MSF.465-466.361.
[49] Zhou Z, Cai X, Zhao Y, Chen L, Xiong C, Li X. Strength characteristics of dry and saturated rock at different strain rates. Trans Nonferrous Met Soc China (English Ed) 2016;26:1919–25. https://doi.org/10.1016/S1003-6326(16)64314-5.
[50] Wong LNY, Li Z, Kang HM, Teh CI. Dynamic loading of carrara marble in a heated state. Rock Mech Rock Eng 2017;50:1487–505. https://doi.org/10.1007/s00603-017-1170-x.
[51] Van Eeckhout EM. The mechanisms of strength reduction due to moisture in coal mine shales. Int J Rock Mech Min Sci Geomech 1976;13:61–7. https://doi.org/10.1016/0148-9062(76)90705-1.

[52] Mcconnell BJ. Factors controlling sandstone strength and deformability in uniaxial compression. Dr Diss Univ Bristol; 1989.
[53] Cai X, Zhou Z, Du X, Zang H. Water-weakening effects on the mechanical behavior of different rock types: phenomena and mechanisms. Appl Sci 2019;9:4450. https://doi.org/10.3390/app9204450.
[54] Zhou Z, Tan L, Cai X. Water infusion on the stability of coal specimen under different static stress conditions. Appl Sci 2020. https://doi.org/10.3390/app10062043.
[55] Haynes WM. Handbook of Chemistry and Physics. CRC Press; 2016. 10.1201/9781315380476.
[56] Violay M, Nielsen S, Spagnuolo E, Cinti D, Di Toro G, Di Stefano G. Pore fluid in experimental calcite-bearing faults: Abrupt weakening and geochemical signature of co-seismic processes. Earth Planet Sci Lett 2013. https://doi.org/10.1016/j.epsl.2012.11.021.
[57] Kameda J, Uno M, Conin M, Ujiie K, Hamada Y, Kimura G. Fault weakening caused by smectite swelling. Earth, Planets Sp 2019. https://doi.org/10.1186/s40623-019-1108-5.
[58] Laird DA. Influence of layer charge on swelling of smectites. Appl Clay Sci 2006. https://doi.org/10.1016/j.clay.2006.01.009.
[59] Atkinson BK. Fracture Mechanics of Rock. Elsevier Ltd. 1987. https://doi.org/10.1016/C2009-0-21691-6.
[60] Jiménez-González I, Rodríguez-Navarro C, Scherer GW. Role of clay minerals in the physicomechanical deterioration of sandstone. J Geophys Res Earth Surf 2008. https://doi.org/10.1029/2007JF000845.
[61] Anderson RL, Ratcliffe I, Greenwell HC, Williams PA, Cliffe S, Coveney PV. Clay swelling - A challenge in the oilfield. Earth-Science Rev 2010. https://doi.org/10.1016/j.earscirev.2009.11.003.
[62] Das BM. Soil mechanics. Eng. Handbook, Second Ed., 2004. 10.1201/b15620-17.
[63] Aben FM, Doan ML, Gratier JP, Renard F. High strain rate deformation of porous sandstone and the asymmetry of earthquake damage in shallow fault zones. Earth Planet Sci Lett 2017;463:81–91. https://doi.org/10.1016/j.epsl.2017.01.016.
[64] Eppes MC, Keanini R. Mechanical weathering and rock erosion by climate-dependent subcritical cracking. Rev Geophys 2017;55:470–508. https://doi.org/10.1002/2017RG000557.
[65] Anderson TL. Fracture Mechanics: Fundamentals and Applications. Fourth Edition; 2017.
[66] Atkinson BK. Subcritical crack growth in geological materials. J Geophys Res Solid Earth 1984;89:4077–114. https://doi.org/10.1029/JB089iB06p04077.
[67] Nara Y, Morimoto K, Hiroyoshi N, Yoneda T, Kaneko K, Benson PM. Influence of relative humidity on fracture toughness of rock: Implications for subcritical crack growth. Int J Solids Struct 2012;49:2471–81. https://doi.org/10.1016/j.ijsolstr.2012.05.009.
[68] Song Z, Frühwirt T, Konietzky H. Inhomogeneous mechanical behaviour of concrete subjected to monotonic and cyclic loading. Int J Fatigue 2020;132:105383https://doi.org/10.1016/j.ijfatigue.2019.105383.
[69] Song Z, Konietzky H, Herbst M. Bonded-particle model-based simulation of artificial rock subjected to cyclic loading. Acta Geotech 2019;14:955–71. https://doi.org/10.1007/s11440-018-0723-9.
[70] Peng K, Zhou J, Zou Q, Yan F. Deformation characteristics of sandstones during cyclic loading and unloading with varying lower limits of stress under different confining pressures. Int J Fatigue 2019. https://doi.org/10.1016/j.ijfatigue.2019.06.007.
[71] Peng K, Wang Y, Zou Q, Liu Z, Mou J. Effect of crack angles on energy characteristics of sandstones under a complex stress path. Eng Fract Mech 2019. https://doi.org/10.1016/j.engfracmech.2019.106577.
[72] Peng K, Zhou J, Zou Q, Song X. Effect of loading frequency on the deformation behaviours of sandstones subjected to cyclic loads and its underlying mechanism. Int J Fatigue 2020. https://doi.org/10.1016/j.ijfatigue.2019.105349.
[73] Dang W, Konietzky H, Frühwirt T, Herbst M. Cyclic Frictional Responses of Planar Joints Under Cyclic Normal Load Conditions: Laboratory Tests and Numerical Simulations. Rock Mech Rock Eng 2020. https://doi.org/10.1007/s00603-019-01910-9.
[74] Griffith AA. The Phenomena of Rupture and Flow in Solids. Philos Trans R Soc A Math Phys Eng Sci 1921;221:163–98. https://doi.org/10.1098/rsta.1921.0006.
[75] Nasseri MHB, Schubnel A, Young RP. Coupled evolutions of fracture toughness and elastic wave velocities at high crack density in thermally treated Westerly granite. Int J Rock Mech Min Sci 2007. https://doi.org/10.1016/j.ijrmms.2006.09.008.
[76] Kachanov M. Elastic Solids with Many Cracks and Related Problems. Adv Appl Mech 1993. https://doi.org/10.1016/S0065-2156(08)70176-5.
[77] Zhai Y, Ma G, Zhao J, Hu C. Dynamic failure analysis on granite under uniaxial impact compressive load. Front Archit Civ Eng China 2008;2:253–60. https://doi.org/10.1007/s11709-008-0042-6.
[78] Paliwal B, Ramesh KT. An interacting micro-crack damage model for failure of brittle materials under compression. J Mech Phys Solids 2008;56:896–923. https://doi.org/10.1016/j.jmps.2007.06.012.
[79] Wu Q, Weng L, Zhao Y, Guo B, Luo T. On the tensile mechanical characteristics of fine-grained granite after heating/cooling treatments with different cooling rates. Eng Geol 2019. https://doi.org/10.1016/j.enggeo.2019.03.014.